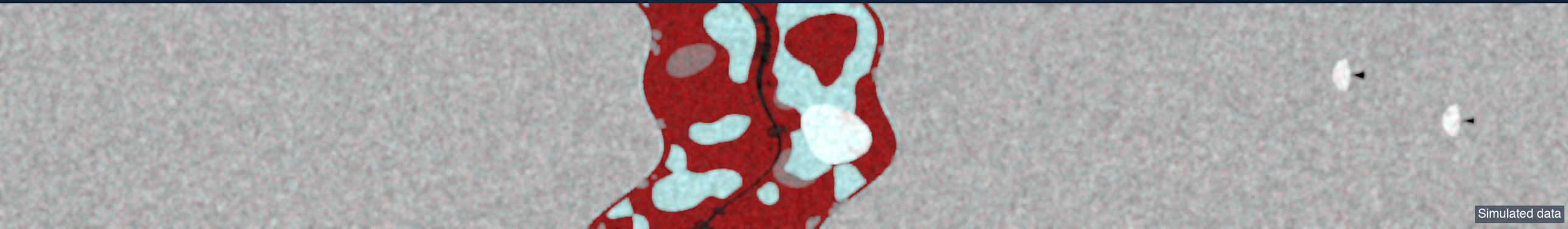
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Flood Mapping with SAR

Seeing floodwater through clouds, day and night

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

Use radar images to map where a flood is, how much land it covers, and how often it happens, even under cloud.

Why water looks dark, how to turn images into flood maps, how to check them, and how engineers use them.

THIS LECTURE

Outline

01 **Floods and why radar sees them**
Flood types, clouds, how water scatters radar, and the satellites

02 **Mapping water in SAR images**
Preparing images, thresholds, change detection and false alarms

03 **Accuracy, services and tools**
Checking a flood map, and the free services and software

04 **Case studies**
Kerala 2018, Pakistan 2022, Valencia 2024 and Assam’s flood atlas

05 **Engineering practice**
Water depth, flood frequency, model calibration, limits and exercises

01

PART 1

Floods and why radar sees them

Flood types, clouds, how water scatters radar, and the satellites

FLOODS

Four kinds of flood

River floods

Rivers overtop their banks for days to weeks, as on the monsoon-fed Brahmaputra.

SAR: well suited

Flash floods

Intense rain on steep or paved ground: water rises and falls within hours.

SAR: often missed

Coastal floods

Storm surges from cyclones, and very high tides, push the sea inland.

SAR: well suited

Urban floods

Rain overwhelms drains; shallow water between buildings.

SAR: difficult

What decides whether SAR can map a flood: how long the water stays, how open the area is, and when the satellite passes.

FLOODS

Why radar for floods

Clouds
Floods come with storms; optical satellites see only cloud tops

Day and night
Radar brings its own illumination

10 m
Pixel spacing of free Sentinel-1 images

Since 2014
Free Sentinel-1 archive for flood history

Water is easy to see

Calm water reflects the radar away, so it appears dark against the land.

Large areas at once

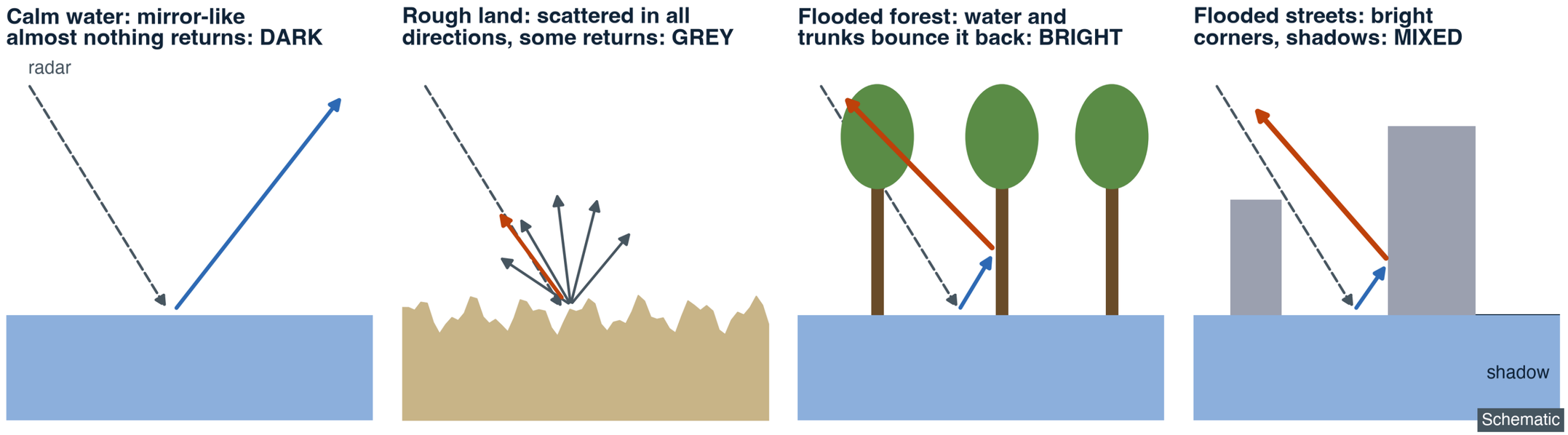
One Sentinel-1 image covers 250 km: a whole river basin or delta.

But timing matters

A satellite passes only every few days; a fast flood may come and go between passes.

FLOODS

Why floodwater looks dark, and when it does not



Schematic. Smooth surfaces act like mirrors; vertical objects standing in water act like corner reflectors.

Dark: open floodwater, the easy case.

Bright: flooded forest and flooded towns, from double bounce.

Grey: water roughened by wind or heavy rain can look like land.

FLOODS

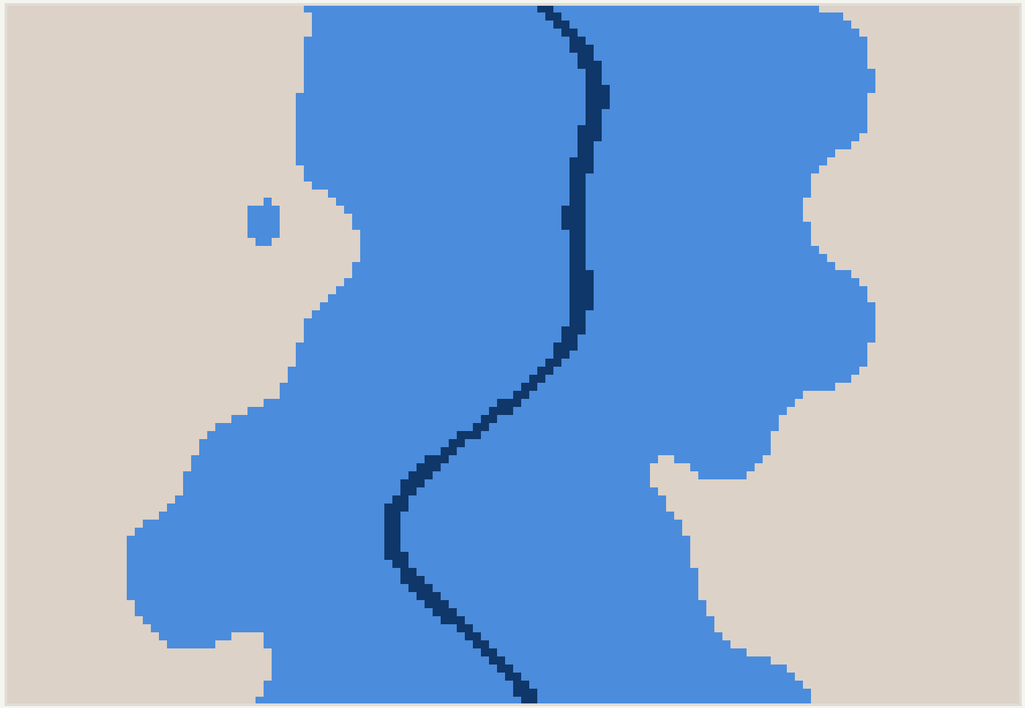
Radar satellites for flood mapping

Satellite	Band	Notes
Sentinel-1 (EU)	C-band	6–12 days; free; the workhorse for flood mapping
NISAR (NASA–ISRO)	L- and S-band	12 days; free; sees under forest canopies
EOS-04 / RISAT-1A (India)	C-band	Launched 2022; used for Indian flood mapping
RADARSAT Constellation (Canada)	C-band	Three satellites; frequent revisits
ALOS-2 and ALOS-4 (Japan)	L-band	Good in forests; tasked for disasters
ICEYE, Capella and others	X-band	Commercial small satellites; images within hours

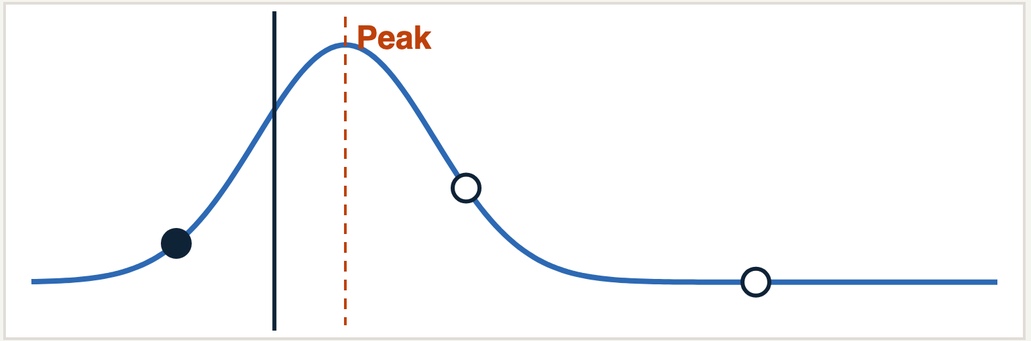
FLOODS

Timing: will the satellite catch the peak?

Day 6 The flood right now



River level and satellite passes (every 6 days)



Pass Pass Pass

What the satellite saw



waiting…

waiting…

Day 3 Day 9 Day 15

No pass caught the peak on day 7

02

PART 2

Mapping water in SAR images

Preparing images, thresholds, change detection and false alarms

MAPPING

Preparing the images

01
Download GRD images (IW, VV + VH)

02
Apply precise orbits

03
Remove thermal noise

04
Calibrate to σ^0 (backscatter)

05
Filter speckle

06
Terrain-correct (geocode)

07
Convert to decibels (dB)

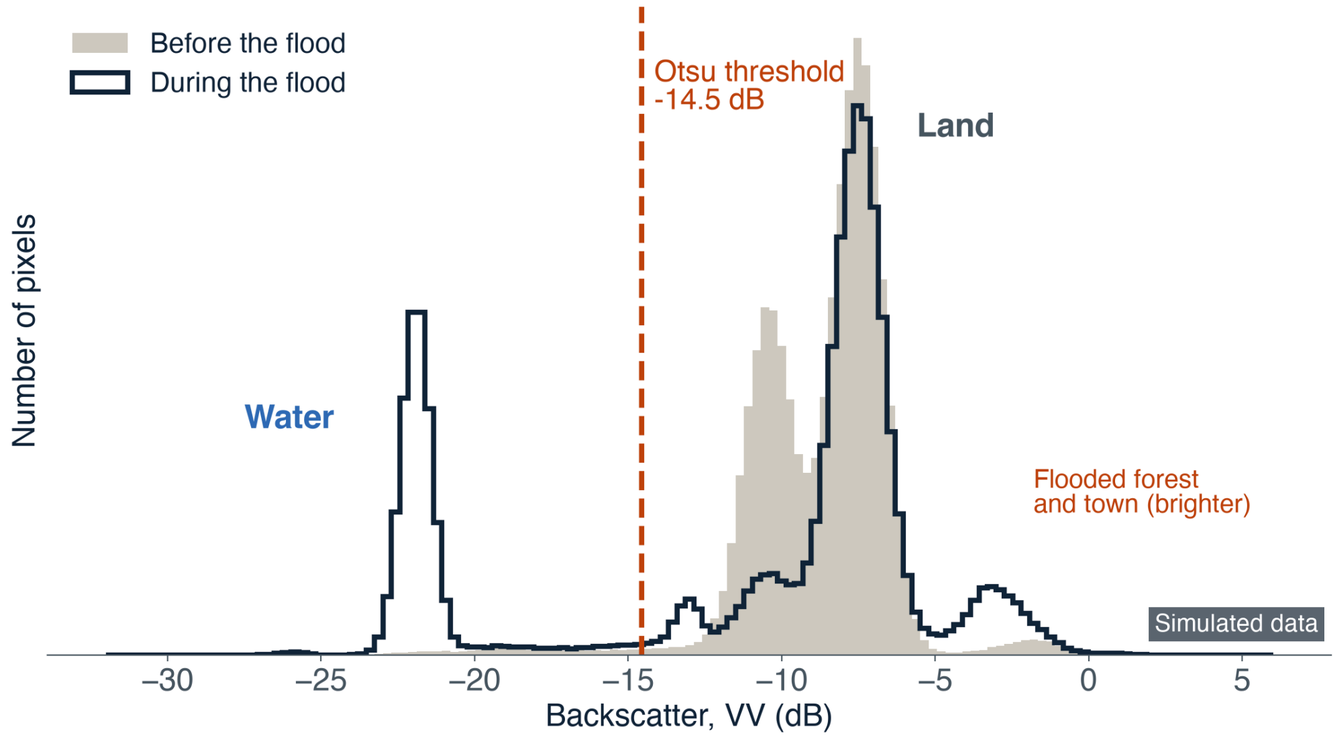
08
Stack before and during

In SNAP: one graph does steps 2–7 for each image.

In Google Earth Engine: Sentinel-1 images are already calibrated and geocoded.

MAPPING

Thresholding: separating water from land



Simulated scene: during the flood a dark water peak appears.

Pixels darker than the threshold are classed as water. Here Otsu's method picks **-14.5 dB** automatically.

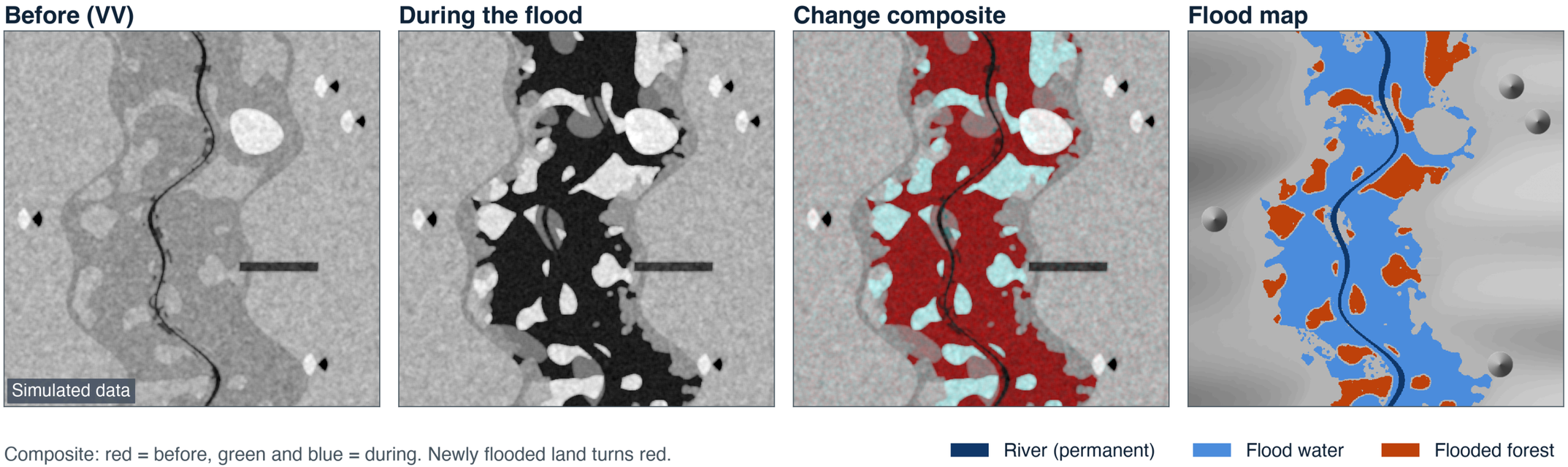
Otsu chooses the value that best splits the histogram into two groups.

The best value changes with wind, incidence angle and scene: compute it for each image, or for tiles.

Fast and simple, but anything dark becomes 'water'.

MAPPING

Change detection: before and during



Composite: red = before, green and blue = during. Newly flooded land turns red.

Simulated scene, 15 km across.

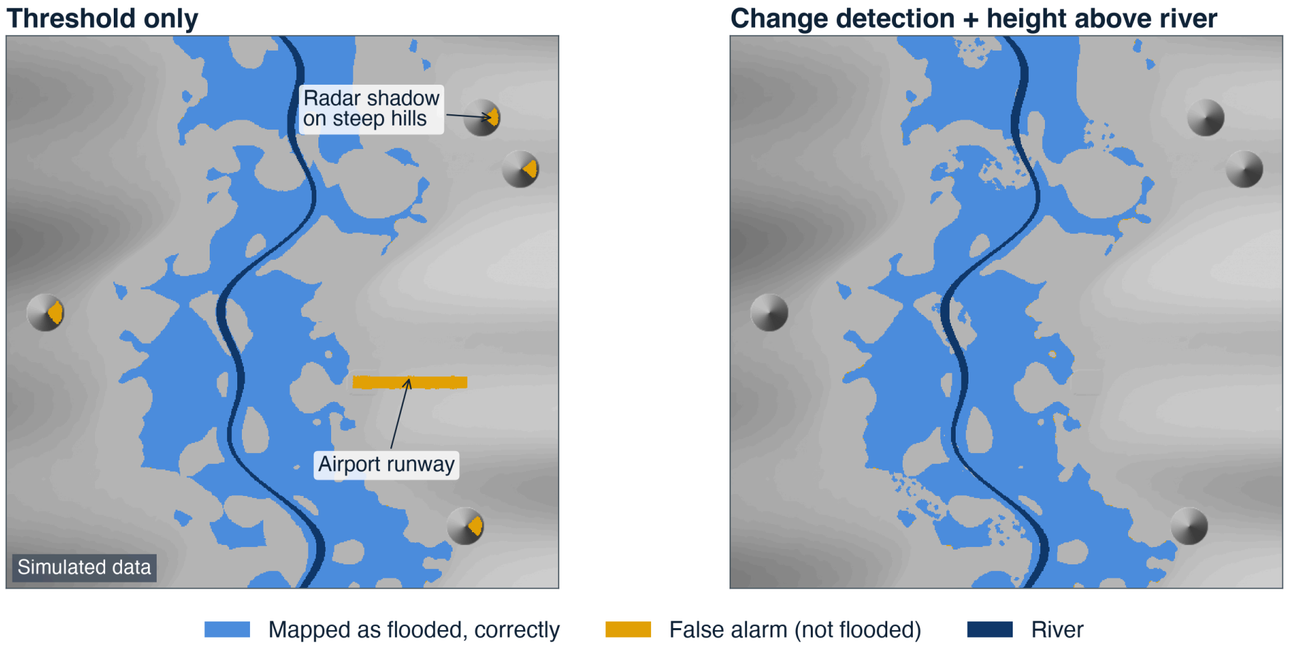
Darker by more than 3 dB: new open water.

Brighter by more than 3 dB in forest: flooded vegetation.

Unchanged: not flooded, even if dark (runways, shadows).

MAPPING

False alarms: dark but not flooded



Simulated: false alarms shown in yellow.

Lookalikes: radar shadow, runways and smooth roads, dry sand, calm reservoirs.

Fix 1: change detection. They were dark before the flood too.

Fix 2: height above the nearest river (HAND). Land 15–20 m above the river cannot flood.

Fix 3: mask permanent water and steep slopes.

MAPPING

The hard cases

Flooded vegetation

Gets brighter, not darker, from double bounce. Look for increases; L-band works best under canopies.

Urban floods

Bright corners and shadows mix with water. Use coherence loss, very high resolution and local knowledge.

Wind and heavy rain

Rough water looks like land and is missed. VH is less affected than VV.

Fast floods

Water may have gone before the next pass. Combine satellites, and use models and field reports.

MAPPING

A flood-mapping workflow

01 Pre-flood reference image

02 Flood-time image

03 Prepare both

04 Change detection

05 Threshold

06 Masks: HAND, river, slope

07 Clean small patches

08 Check and share

Reference: same track and pass, same season, dry conditions.

Output: a water mask, an area in km², and a map people can read.

03

PART 3

Accuracy, services and tools

Checking a flood map, and the free services and software

ACCURACY

How good is each method?

Method (simulated scene)	Flood found	False alarms	CSI
Threshold only	65%	3%	0.64
Change detection	68%	0%	0.68
Change detection + HAND	68%	0%	0.67
+ rule for flooded forest	87%	0%	0.87

Simulated scene: 78 km² truly flooded (river excluded).

Thresholding alone also maps runways and shadows.

Change detection removes them, but misses flooded forest.

Adding the brightening rule finds most of the rest.

ACCURACY

Checking a flood map

CSI = hits ÷ (hits + misses + false alarms)

Hit rate

Share of real flooding that was mapped

False alarm ratio

Share of the mapped flood that was dry

Reference data

Cloud-free optical images (Sentinel-2), aerial and drone photos, field reports and high-water marks, all from the same day if possible.

SERVICES

Who makes flood maps in a disaster

Copernicus Global Flood Monitoring

Since 2021: every Sentinel-1 image of land is mapped automatically by three algorithms, usually within about five hours.

Copernicus EMS Rapid Mapping

Detailed maps on request during an emergency, from radar, optical and aerial images.

International Charter

Space agencies share satellite images free of charge for major disasters.

NRSC (ISRO), India

Flood maps for Indian states during the monsoon, shared through NDEM and Bhuvan.

TOOLS

Free software for flood mapping

Google Earth Engine

UN-SPIDER’s ready-made Sentinel-1 flood script: change detection plus exposure maps.

ESA SNAP

Prepare images, threshold, and export, all in one graph.

QGIS

Map the result, measure flooded areas, overlay roads and villages.

Python

rasterio for images, scikit-image for Otsu thresholds, geopandas for exposure.

HEC-RAS

2D flood models, checked against SAR flood maps (Part 5).

GFM data

Download ready-made Sentinel-1 flood extents for any past date.

04

PART 4

Case studies

Kerala 2018, Pakistan 2022, Valencia 2024 and Assam’s flood atlas

CASE STUDY

Kerala, India, August 2018

36%
More rain than normal, 1 June to 29 August 2018

≈ 520 km²
Flooded at the peak, from SAR images

400+
People killed in the floods and landslides

17
Satellite datasets NRSC used for about 20 flood maps

Weeks of monsoon cloud: radar gave the only clear view.

NRSC mapped flooding from RADARSAT-2 and Sentinel-1 within days.

Paddy and coconut groves: flooded vegetation made mapping harder.

Sources: NRSC (ISRO) flood maps; Geomatics, Natural Hazards and Risk (2019); PLOS ONE (2020).

CASE STUDY

Pakistan, 2022

≈ 30,500 km²
Seen flooded at least once in Sentinel-1, 10 Aug–23 Sep

30 Aug
Day of the largest flooded area

33 million
People living in or near the flooded areas

≈ 85,000 km²
Land ‘affected by flood waters’ in a separate UNOSAT estimate

A slow monsoon flood across the Indus plains: ideal for radar.

Repeated Sentinel-1 images showed how the water spread and drained.

Numbers differ with the definition: open water seen versus land affected.

Sources: Roth et al. (2023), Natural Hazards and Earth System Sciences; UNOSAT.

CASE STUDY

Valencia, Spain, October 2024

29 Oct
Flash flood after extreme rain from a cut-off low (DANA)

> 220
People killed

EMSR773
Copernicus Emergency Management Service activation

140,000+
Vehicles destroyed or disabled

Water rose and drained within hours, between satellite passes.

Streets were left full of mud and cars rather than open water.

Rapid mapping combined radar, optical and aerial images.

Sources: Copernicus EMS; European Geographical Journal (2024).

CASE STUDY

Assam, India: a flood atlas

26 years
Of satellite images, 1998 to 2023

400+
Optical and radar images of floods

3rd
Edition of the Flood Hazard Zonation Atlas

Village
Level of detail for how often land floods

Brahmaputra floods come every monsoon: the question is how often each place floods.

Frequency maps guide embankments, relief planning and land use.

Built by NRSC (ISRO) for the Assam State Disaster Management Authority.

Source: NRSC (ISRO), Flood Hazard Zonation Atlas of Assam, 1998–2023.

05

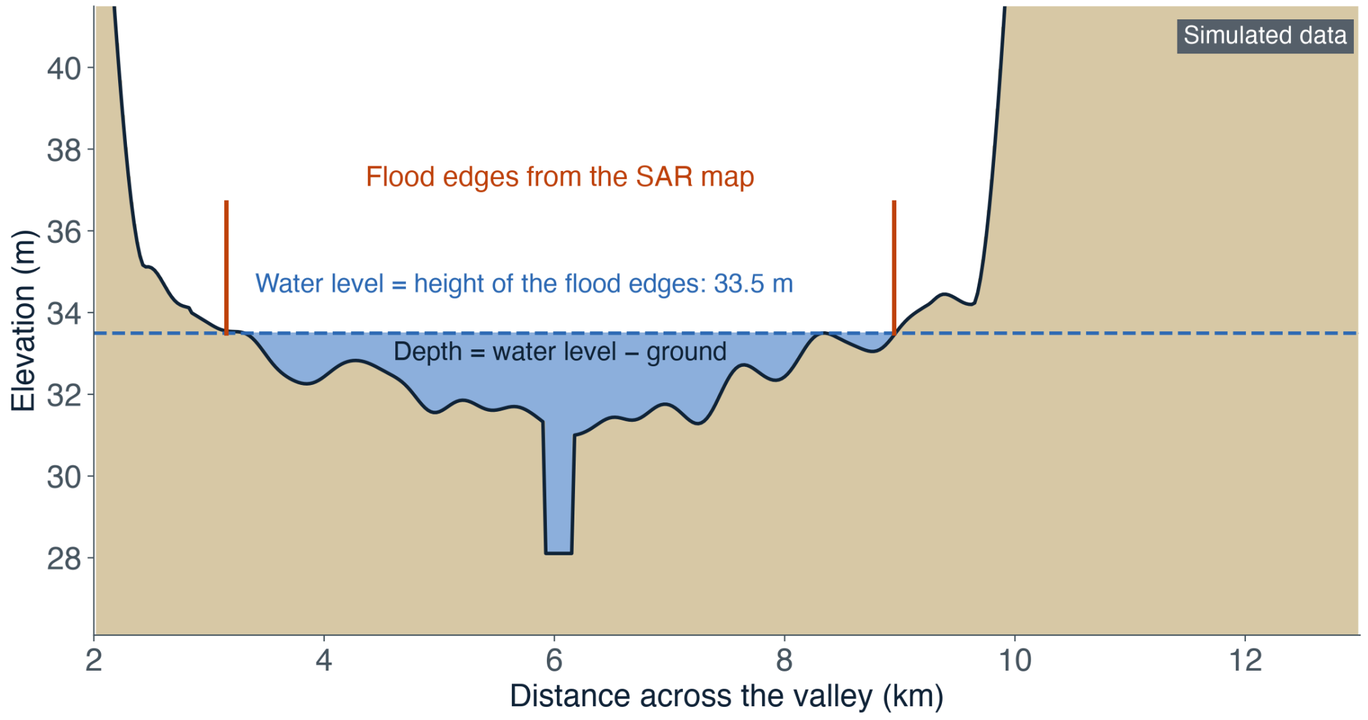
PART 5

Engineering practice

Water depth, flood frequency, model calibration, limits and exercises

ENGINEERING

From flood extent to water depth



Simulated cross-section through the scene in Part 2.

SAR gives the flood **edges**, not the depth.

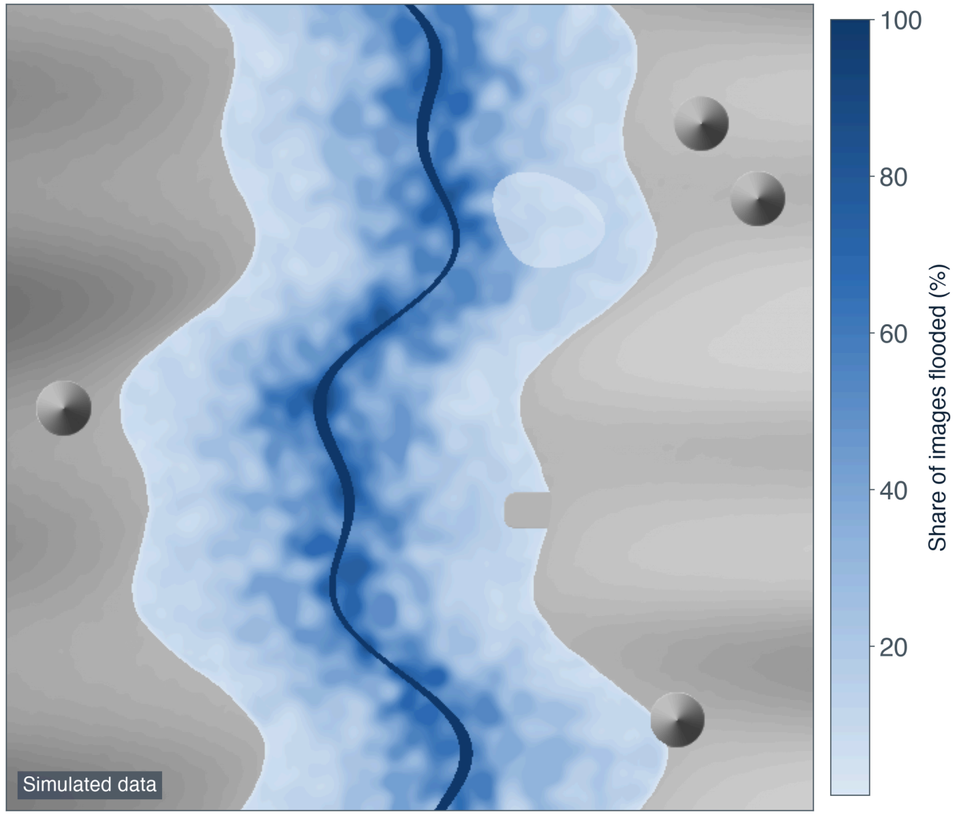
The ground height at the edges gives the water level: here 33.5 m.

Depth = water level – ground: up to about 2.5 m on this floodplain.

Needs an accurate elevation model; errors of a metre matter.

ENGINEERING

How often does each place flood?



Simulated: 240 images over 10 years.

Stack every flood map in the archive and count how often each pixel was wet.

Dark blue floods most years; pale blue only in big floods.

Engineers use it for zoning, embankment and road levels, and insurance.

Sentinel-1 since 2014 gives about ten years of history.

ENGINEERING

Calibrating flood models with SAR

The model

A 2D hydraulic model (HEC-RAS, LISFLOOD-FP) predicts where water goes for a given flow.

The adjustment

Change roughness (Manning’s n) and boundary conditions until the extents agree.

The check

Compare the modelled extent with the SAR flood map on the same day; score it with CSI.

The result

A calibrated model can then map floods the satellite never saw, including the 100-year flood.

ENGINEERING

Where engineers use SAR flood maps

Emergency response

Where to send boats, food and medical teams; which roads are cut.

Damage and exposure

Crops, roads, buildings and people inside the flood.

Design levels

Embankment heights, bridge clearances and road levels.

Planning and zoning

Keep new building out of land that floods often.

Insurance and finance

Payouts triggered by mapped flood extent.

Forecast checks

Was the forecast flood the real one?

ENGINEERING

Limits and good practice

Limits

- A snapshot every few days: may miss the peak
- Flooded forest and towns are hard
- Wind and rain can hide water
- Extent only: depth needs a good DEM

Good practice

- Use change detection, not just a threshold
- Mask with HAND, slope and permanent water
- Combine satellites, optical images and field reports
- State the date, time and method on every map

EXERCISES

Try these

01 A pixel’s backscatter is 0.01 in linear units. What is it in dB? And 0.1?

02 A field measures −9 dB before a flood and −21 dB during it. What is the change, and is it flooded by the 3 dB rule?

03 A flood map has 1,250,000 flooded pixels at 10 m spacing. What area is that in km²?

04 Compared with drone photos, a map has 800 hits, 100 false alarms and 200 misses. Find the hit rate, the false alarm ratio and the CSI.

EXERCISES

Answers

01 $10 \log_{10}(0.01) =$ **−20 dB**; $10 \log_{10}(0.1) =$ **−10 dB**. Each factor of 10 is 10 dB.

02 −21 − (−9) = **−12 dB**: much darker than the 3 dB rule, so flooded.

03 1,250,000 × 100 m² = 125,000,000 m² = **125 km²**.

04 Hit rate 800 ÷ 1,000 = **80%**; false alarms 100 ÷ 900 ≈ **11%**; CSI 800 ÷ 1,100 ≈ **0.73**.

RECAP

Key points

01 Radar maps floods through cloud, day and night.

02 Calm water is dark; flooded forest and towns can be bright.

03 Change detection plus masks beats a single threshold.

04 Check maps against independent data, and always give the date.

05 Extent plus elevation gives depth; the archive gives flood frequency.

FURTHER READING

References

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Thank you. Questions?

Figures and the animation were made for this lecture. The radar scene, flood maps, accuracy scores, depth profile and frequency map are simulated, with Sentinel-1-like speckle and backscatter, and labelled as such; the scattering diagram is schematic. Case-study facts: NRSC (ISRO) and published studies (Kerala); Roth et al. (2023) and UNOSAT (Pakistan); Copernicus EMS (Valencia); NRSC Flood Hazard Zonation Atlas (Assam). Services: Copernicus Emergency Management Service, International Charter, UN-SPIDER. Sentinel-1 data: EU Copernicus programme / ESA. Full references on the previous slide.